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EX NO: 11	Implementation of IoT-based Smart Agriculture using MATLAB
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AIM: To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

Procedure:

1. Define soil moisture range (0–100%) and set an irrigation threshold value (e.g., 40%).
2. Generate or input soil moisture data to simulate field conditions.
3. Compare soil moisture with the threshold value to determine pump status (ON if low moisture, OFF if sufficient moisture).
4. Simulate water pumping action based on control logic.
5. Calculate irrigation efficiency based on moisture improvement and water usage.
6. Display soil moisture level, pump status, and efficiency results.
7. Plot graphs to analyze soil moisture variation and irrigation performance.
8. Evaluate system behavior for smart agriculture decision-making.

Objective 1 Matlab Code and Output:

```
clc; clear; close all;
```

```
% Simulated soil moisture data (%)
```

```
time = 1:10;
```

```
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
```

```
% Threshold for irrigation
```

```
threshold = 40;
```

```
figure
```

```
% ----- Graph 1: Soil Moisture Variation -----
```

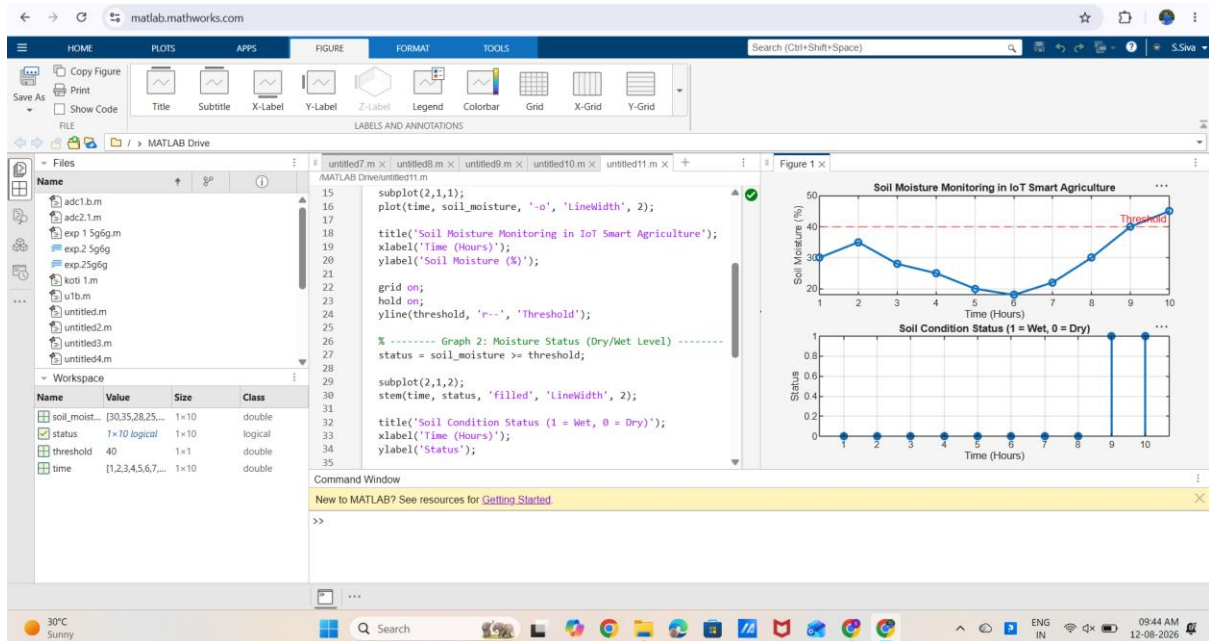
```
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')

grid on

hold on
yline(threshold,'r--','Threshold')
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
status = soil_moisture >= threshold;
subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')

grid on
```

OUTPUT:



Result:

Soil moisture levels vary over time, showing fluctuations in field water content.

The threshold line helps identify dry and wet conditions for smart irrigation decisions.